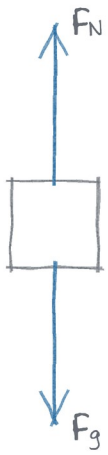
Free-Body Diagrams

1. WARM-UP

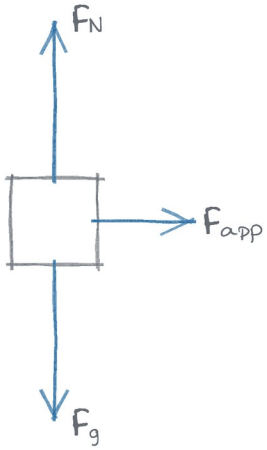
A box of books sits at rest on a level floor. Draw the free-body diagram for the box.



Normal force equals weight here only because nothing else pushes up or down.

2. WARM-UP

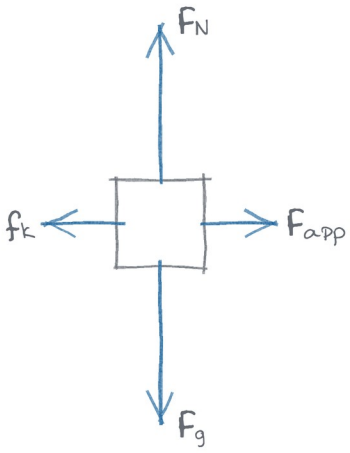
A crate on smooth, frictionless ice is pushed to the right and speeds up. Ignore air resistance. Draw the free-body diagram for the crate.



Fapp is the only sideways force. No friction arrow and no arrow for motion.

3. **WARM-UP**

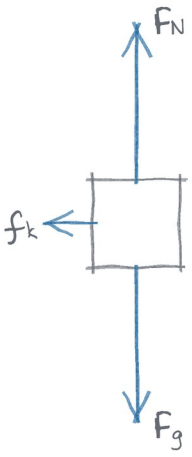
A dresser is pushed to the right across a rough wooden floor at a constant speed. Ignore air resistance. Draw the free-body diagram for the dresser.



Constant speed means the push and the friction are equal. The push is not bigger.

4. **PRACTICE**

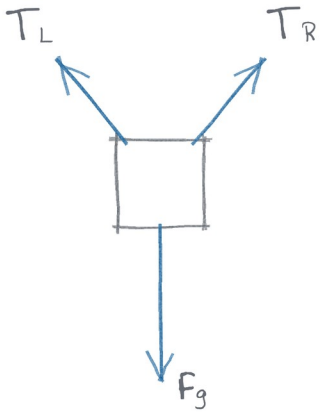
A hockey puck slides to the right across rough ice and slows to a stop. Nothing is pushing it. Ignore air resistance. Draw the free-body diagram for the puck while it is still sliding.



No forward force. Nothing pushes the puck once it is sliding.

5. **PRACTICE**

A framed picture hangs at rest from two ropes. One rope pulls up and to the left and the other pulls up and to the right, each at 50° above the horizontal. Draw the free-body diagram for the picture.



Two T arrows, one along each rope, both the same length.

6. **PRACTICE**

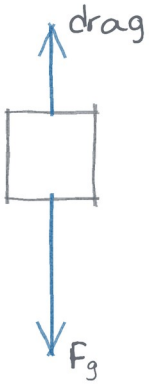
A person stands on a bathroom scale in an elevator. The elevator is moving up and speeding up. Draw the free-body diagram for the person.



Speeding up while going up, so the normal force is longer than the weight.

7. **PRACTICE**

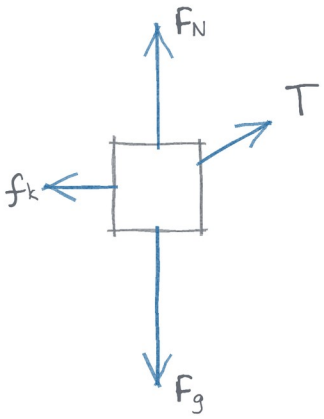
A skydiver falls toward the ground and is still speeding up. Air resistance acts on her. Draw the free-body diagram for the skydiver.



Still speeding up, so drag is shorter than gravity.

8. **CHALLENGE**

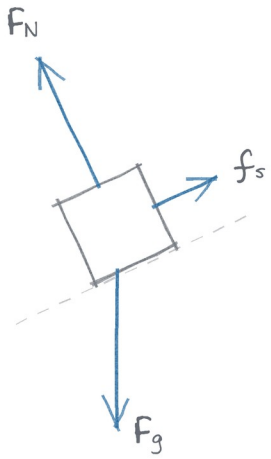
A sled is pulled to the right across rough snow at a constant speed. The rope pulls at 30° above the horizontal. Ignore air resistance. Draw the free-body diagram for the sled.



The rope lifts a little, so the normal force is less than the weight.

9. **CHALLENGE**

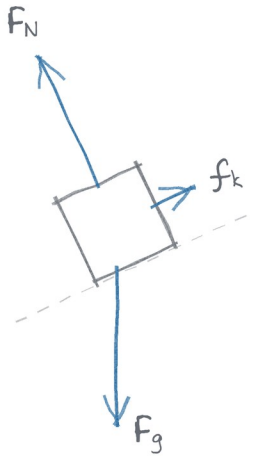
A toolbox sits at rest on a rough ramp tilted 25° above the horizontal. Draw the free-body diagram for the toolbox.



Normal force is perpendicular to the ramp. Gravity points straight down, not into the ramp.

10. **CHALLENGE**

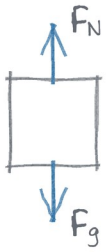
A wooden block slides down a rough ramp tilted 25° above the horizontal and speeds up as it goes. Ignore air resistance. Draw the free-body diagram for the block.



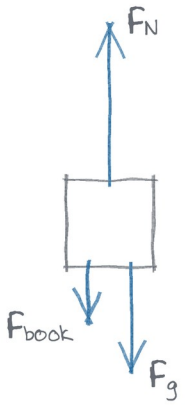
Friction points up the ramp, opposite the sliding.

11. **CHALLENGE**

A book rests on a table, and the table stands on a level floor. Nothing is moving. Draw two separate free-body diagrams: one for the book and one for the table.



Book

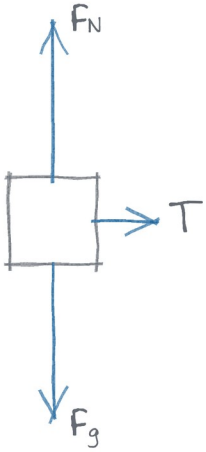


Table

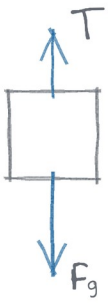
The book's weight is not drawn on the table. The table feels the book's push, Fbook.

12. **MULTI-TOPIC**

A block on a frictionless table is tied to a string. The string runs over a pulley at the right edge of the table to a hanging weight, and the weight falls and speeds up. Ignore the mass of the string and the pulley, and ignore air resistance. Draw two separate free-body diagrams: one for the block and one for the hanging weight.



Block



Hanging weight

Gravity on the weight is longer than the tension, because the weight speeds up downward.